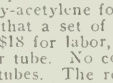
when tubes are prossered. It is unnecessary to give any special attention to welded tubes when boilers are washed. Reports received from different roads indicate that considerable difficulty was experienced in welding flues with the oxy-acetylene process, while the roads using the electric process report very satisfactory results, especially one of the roads that tried out both methods and adopted the electric process. Therefore, the committee recommends electric process for welding flues, and also recommends that flues be set with coppers, prossered and beaded in regular manner before welding, until it is clearly demonstrated that other methods are equally as good or better.

One road using oxy-acetylene for welding the flues reports that it costs, including gas and all work preparing flue sheets and safe-ending flues, an average of 71 cents per flue, and the regular method, which consists of safe-ending and applying coppers and other work for the same number of flues, is 22 cents per flue.

A road using oxy-acetylene for setting the flues, as shown in Fig. 1, reports that a set of welded flues, containing 335 flues, amounts to $18 for labor, $15 for gas, making a total of $33, or $.098 per tube. No coppers are used by this road when welding in tubes. The regular method for the same number of flues costs $25.25, or $.075 per flue. These figures include only such work as necessary to set flues.

A road using the electric process of welding flues reports that the cost of labor complete to weld beads is $.037 per flue. The cost to weld in new flues is $.03 per flue.

A road using the electric process for welding the flues as shown in Fig. 2 reports the cost of welding 34 4½-in.



ELECTRIC WELDED

Fig. 4—Method of Welding Door Sheets to Mud Ring.

tubes and 32 2-in. tubes—labor, $4.12; current, $2; material, $1.25. Total, $7.37. One 4½-in. tube is equivalent to two 2-in. tubes; therefore, this amounts to 100 2-in. tubes costing $.073 per flue. The standard method of setting the flues by the same road, which consists of safe-ending, applying coppers and prossering, is $.089 per flue, and adding the cost of welding ($.073) makes a total of $.162, which is the cost of welding one 2-in. tube, as shown by Fig. 2.

Below is the cost of setting flues by different roads. This cost applies to the old method:

A. $.06 a flue ready for testing, new flues.
B. $.06 a flue ready for testing, new flues.
B. $.14 removing, cleaning, safe-ending and resetting.
C. $.13 removing, cleaning, safe-ending and resetting.
D. $.067 installing copper ferrules, setting flues, rolling front ends, setting and beading firebox end of tube.
E. $.044 per flue for setting tubes only, includes copper ferrules.
F. $.04 per flue for setting tubes only, includes copper ferrules.
G. $.036 per flue for setting tubes only, includes copper ferrules.

From these figures, it appears that the cost of welding flues in accordance with electric process is cheaper than the oxy-acetylene process. It also gives better service. None of the roads are welding-in tubes in the front tube sheet, as it does not seem to be desirable to change the method now used.

Vertical cracks in the side sheets are welded with both oxy-acetylene and electric process. Seven roads get good results with electric process when cracks do not exceed 12 in. in length. Two roads advised very poor success with welding vertical cracks with the oxy-acetylene process. The method used by all railroads is to V out the sheet ⅜ in. to ½ in. at the top and ⅛ in. to 3/16 in. at the bottom.

Cracks in the top of the back flue sheet as long as 38 in. have been welded successfully with both types of welder. The sheet is cut out a few inches beyond the end of the crack to avoid a new crack from starting and is dressed out V shape the same as the cracks in the firebox. Some roads preheat the sheets before welding.

Cracks are successfully welded in the crown sheet with both processes. One road, using oxy-acetylene process for welding cracks in the flue sheet, cut off the forward part of the crown sheet and welded a strip the full width of the crown sheet, about 6 or 8 in. deep. This was done to avoid any new cracks developing at this location. It was found that this gives most serviceable results.

Cracks in the door sheet have been successfully welded with both types, but the oxy-acetylene is preferred, as it gives more satisfactory results. One road preheated sheets before welding.

In preparing firebox sheets for electric welding, the crack is V'd out to an angle of about 65 deg. and the metal is filed bright. For oxy-acetylene it is V'd out to an angle of about 90 degs. and no filing is required; the metal is built up from the fire side of the sheet.

Considerable difficulty has been experienced by roads using both oxy-acetylene and electric welding on account of welded cracks opening up after service from three to eight months. The percentage in some cases, which are few, runs as high as 15 per cent. About 5 per cent. is the average.

It is considered that very good service is obtained in welding cracks and sheets in fireboxes by both processes. Most of the difficulty was experienced in the earlier days when this method was new. The same can be said of new cracks developing adjacent to welded cracks. This trouble is very rare and it is not serious.

Some roads using oxy-acetylene use water on each side of the weld, and others preheat the sheets to take care of the expansion and contraction. Roads using electric welders do not have any special method of taking care of the expansion and contraction.

When welding cracks in the firebox sheets which embrace three or four staybolts, the best practice is to remove the staybolts, whether broken or not, and weld the sheets solid and drill new holes.

When renewing broken staybolts in the vicinity of cracks in the firebox sheet, which have been welded, occasionally a weld will open up due to vibration of hammers used in chipping, calking, or driving up staybolts. These cases are rare, however, and the second weld made usually holds good. To prevent this, some roads endeavor to keep the sheets hot while driving bolts.

Bridges have been welded in both front and back tube sheets with both processes and give fairly good results, although some roads have had difficulties with oxy-acetylene. Sheets in some cases must be preheated before and at the time of welding to take care of expansion and contraction, also to prevent cracking of adjacent bridges. Flue holes are rolled while the sheet is hot.

Full firebox sheets are welded successfully with both types of welders, and in some cases experiments have been made with welding the flange of the flue sheet and the door sheet to the side sheet, omitting the rivets.

One road had a switch engine built at a locomotive works, and all the firebox seams were welded with the oxy-acetylene process. There are no rivets in this firebox except the ones that fasten the box to the mud ring. Engine went into service August, 1912, and has given no trouble with the seams leaking, or otherwise. Three-quarter door sheets have been welded in by one road and the rivets have been omitted where the sheet is welded to the side sheet. Another road

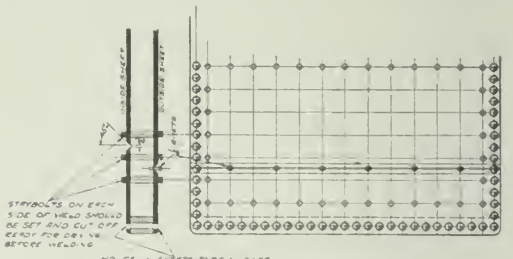


Fig. 5—Method of Applying New Strips Around Bottom of Side Sheets, Throat Sheets and Back Head by Oxy-Acetylene Welding.

reports having welded door sheets from the firebox door down to the mud ring. The rivets were cut out and plugs screwed into rivet holes which were cut off flush with the sheet and welded over as shown in Fig. 4. Fig. 5 shows method of applying new strips around bottom of the side sheet, throat sheet and back head.

One road welded a patch in the side sheet by oxy-acetylene, July, 1909, which was removed March 1, 1913, having made a mileage of 136,980 miles.

Another road, using the electric process, reports that some patches in firebox side sheet fail in from one to three months. Patches in the crown sheet give no trouble. Door-hole collar gives good service, and any patches in the firebox, applied round or oval, will give better service than square ones which are welded on three or four sides.

Another road, to overcome the difficulty they were having with the vertical welding, applied patches, cut diagonally,